

Ankur Desai

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EDUCATION

Michigan State University

B.S. in Computer Science

Expected, May 2028

East Lansing, MI

- **Relevant Coursework:** Linear Algebra, Discrete Mathematics, Object-Oriented Software Design, Computer Organization and Architecture, Data Structures & Algorithms

EXPERIENCE

Claude Builder Club @ MSU

Vice President & Co-Founder

Oct. 2025 - Present

East Lansing, MI

- Engineered serverless automation pipeline integrating GitHub API and Slack SDK to provision team access and workspace invitations, reducing manual onboarding from 15 minutes to <30 seconds for 120+ members.
- Created QR code generation service with AWS S3 storage, CloudFront CDN distribution, and time-based presigned URLs, achieving <100ms global delivery latency with automatic expiration for security.
- Established open source development workflow with GitHub Actions CI/CD, enforcing 85% code coverage via automated test suites and strict linting standards, maintaining comprehensive documentation and contributor guidelines to enable community contributions.

MindMosaic

Software Engineer

Aug. 2025 – Nov. 2025

East Lansing, MI

- Implemented OAuth 2.0 authentication with Google and GitHub providers, enforcing 15-minute access token and 7-day refresh token rotation with httpOnly cookie session management to prevent XSS attacks.
- Architected attribute-based access control (ABAC) with PostgreSQL-backed permission store supporting 3-tier role hierarchy (admin/member/viewer), implementing middleware-based authorization checks.
- Integrated PostgreSQL with Neo4j, adding compound B-tree indexes (user_id, created_at) to reduce note search latency from 850ms to 440ms and cut database CPU utilization by 30% under 50 concurrent queries.

PROJECTS

Mira | *Next.js, Electron, FastAPI, PostgreSQL, PyTorch, Sentence-Transformers*

May 2025 – Nov. 2025

Multi-client intelligence system, analyzing real-time audio streams to execute context-aware automation

- Architected containerized microservice deployment with Docker Compose, orchestrating FastAPI backend, PostgreSQL database, and Redis cache across isolated containers with shared Docker network and persistent volume mounts, integrating Electron desktop client with bi-directional IPC for real-time audio streaming.
- Built production hybrid search pipeline merging BM25 keyword matching with Sentence-BERT embeddings (384-dim), implementing 5-minute conversation boundary detection across multi-stream audio ingestion to optimize context retrieval accuracy for RAG pipeline.
- Developed RESTful API with FastAPI using Pydantic schema validation and OpenAPI 3.0 documentation, implementing lazy-loaded dependency injection for heavy ML models and centralized logging with custom exception middleware for consistent error handling.

WizViz — SpartaHack X | *Python, MediaPipe, OpenCV, NumPy, TensorFlow*

Feb. 2025

Won Interactive Media Track with real-time computer vision game utilizing pose-estimation and gesture-detection

- Designed low-latency gesture pipeline using MediaPipe Pose to map 33-point skeletal landmarks to game actions, achieving 60 FPS with sub-20ms input lag by optimizing the inference loop.
- Created custom vector-based physics engine with AABB collision detection to map 3D real-world movements into 2D game state in real-time.

TECHNICAL SKILLS

Languages Python, C++, Java, JavaScript/TypeScript, SQL, Bash

Backend Frameworks FastAPI, SQLAlchemy, Redis, Alembic, Node.js

Frontend Development React, Next.js, Tailwind CSS, Electron, Vite

Cloud & Infrastructure AWS (EC2, RDS, S3), Docker, GitHub Actions, PostgreSQL, Neo4j